Second Part of our outline

In this part we study about conic section (Parabolas, ellipse and hyperbola)

Conic Sections

① **Parobolas**: A set that consists of all the points in a plane equidistant from a given fixed point and a given fixed line in the plane is a parabola.

Standard equation

① $y^2 = 4ax$ ② $y^2 = -4ax$

③ $x^2 = 4ay$ ④ $x^2 = -4ay$

'a' is distance from origin to focus.

General equations of Parabolas

① $y = a(x-h)^2 + k$ regular form

② $x = a(y-k)^2 + h$ Sidewise form

Q: $2y^2 + 3y - 4x - 3 = 0$

The given equation can be written as:

$$\left(y + \frac{3}{4}\right)^2 = 2\left(x + \frac{33}{32}\right)$$

which is the form of $y^2 = 4ax$

① $y^2 = 12x$

$x = \frac{y^2}{12}$

Now

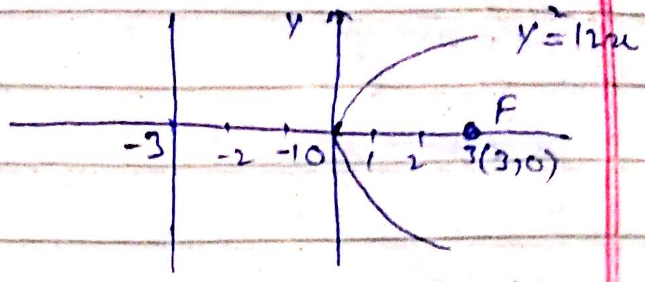
$y^2 = 4ax$

$\cancel{y^2} = 4a\left(\frac{\cancel{y^2}}{\cancel{12}_{3}}\right)$

$\underline{a = 3}$

Focus = $(\pm 3, 0)$

directric is $x = -3$



Graph

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② $y^2 = 16x$

$x = y^2/16$

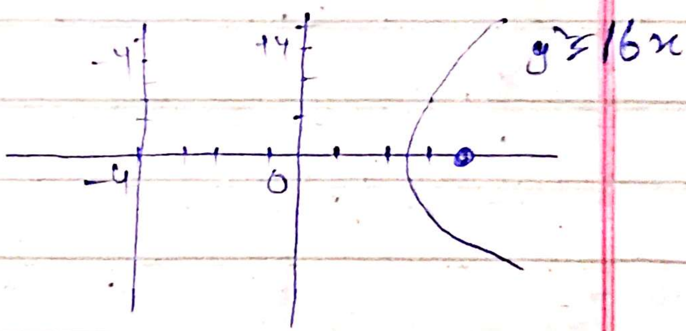
Now

$y^2 = 4ax$

$\cancel{y^2} = 4a(\cancel{y^2}/\cancel{16})$ 4

$a = 4$

Focus = $(4, 0)$



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③ $x^2 = 6y$

$y = \frac{x^2}{6}$

Now

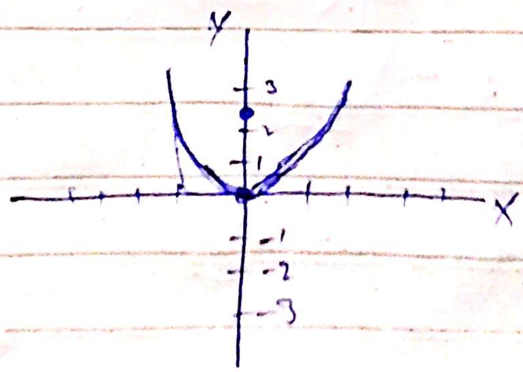
$\cancel{x^2 = 4ay}$

$x^2 = 4a\left(\frac{x^2}{6}\right)$

$\cancel{x^2} = \cancel{4}_{2}a\left(\frac{\cancel{x^2}}{\cancel{6}_{3}}\right)$

$\boxed{a = 3/2}$

Focus $(0, \pm 3/2)$



(4) $x^2 = -12y$

$y = \frac{x^2}{-12}$

$x^2 = -4a \frac{x^2}{-12}$

$\cancel{x^2} = -\cancel{4}a \frac{\cancel{x^2}}{-\cancel{12}_3}$

$a = -3$

Focus $(0, -3)$

(5) $y^2 = -4x$

$x = \frac{y^2}{-4}$

$y^2 = 4a \frac{y^2}{-4}$

$\cancel{y^2} = \cancel{4}a \frac{\cancel{y^2}}{-\cancel{4}_1}$

$a = 1$

Focus $= (1, 0)$

(5) $y^2 = -4x$

$x = \frac{y^2}{-4}$

$\cancel{y^2} = \cancel{4}a \left(\frac{\cancel{y^2}}{-\cancel{4}}\right)$

$\boxed{a = -1}$

Focus $= (-1, 0)$

(6) $y = 4x^2$

$x^2 = 4ay$

$\cancel{x^2} = 4a(4\cancel{x^2})$

$1 = 16a$

$\boxed{a = 1/16}$

Focus $= (0, 1/16)$

(7)

$y = -8x^2$

$\cancel{x^2} = 4a(-8\cancel{x^2})$

$a = -1/32$

Focus $= (0, -1/32)$

(9) $x^2 = y^2$

$y^2 = 4ax$

$\cancel{y^2} = 4a(\cancel{y^2})$

$\boxed{a = 1/4}$

② **Ellipse** : An ellipse is a set of points in a plane whose distances from two fixed points in the plane have a constant sum. ~~This~~ The fixed points are the foci of the ellipse.

Standard equations

⇒ Focus on the x-axis:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a>b)$$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci = $(\pm c, 0)$

Vertices: $(\pm a, 0)$

⇒ Foci on the y-axis:- $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \ (a>b)$

Center-to-focus distance: $c = \sqrt{a^2 - b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

In the these case 'a' is the semimajor axis and 'b' is seminor axis.

Example:- The ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$

Semimajor axis : $a = \sqrt{16} = 4$

Semiminor axis : $b = \sqrt{9} = 3$

Center to focus distance $= c = \sqrt{a^2 - b^2}$

$c = \sqrt{4^2 - 3^2}$

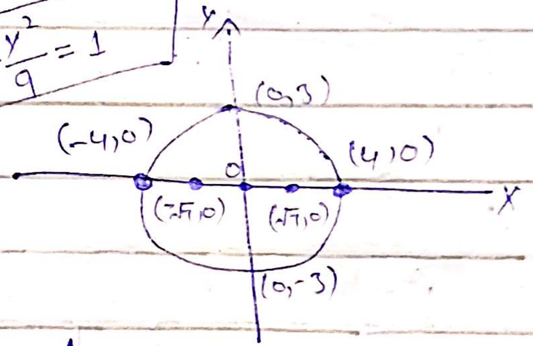
$c = \sqrt{16 - 9} = \sqrt{7}$

Foci: $(\pm c, 0) = (\pm\sqrt{7}, 0)$

Vertices: $(\pm a, 0) = (\pm 4, 0)$

Center: $(0,0)$

$$\boxed{\frac{x^2}{16} + \frac{y^2}{9} = 1}$$

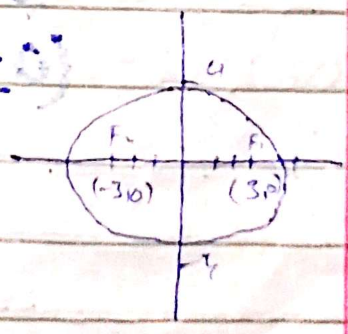


Some questions from 11.6 from 14edition

(17) $16x^2 + 25y^2 = 400$

$$\frac{\cancel{16}x^2}{\cancel{400}_{25}} + \frac{\cancel{25}y^2}{\cancel{400}_{16}} = 1$$

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$



$c = \sqrt{a^2 - b^2} = \sqrt{25 - 16} = \sqrt{9} = 3$

$c = (\pm 3, 0)$

(18) $5x^2+9y^2=45$

$$\frac{\cancel{5}x^2}{\cancel{45}9}+\frac{\cancel{9}y^2}{\cancel{45}5}=1$$

$$\frac{x^2}{9}+\frac{y^2}{5}=1$$

$$c=\sqrt{a^2-b^2}=\sqrt{(3)^2+(\sqrt{5})^2}=\sqrt{9+5}$$
$$=\sqrt{14}$$

$c=(\pm\sqrt{14},0)$

vertices $=(\pm 3,0)$

Center $=(0,0)$

(19)

$2x^2+y^2=2$

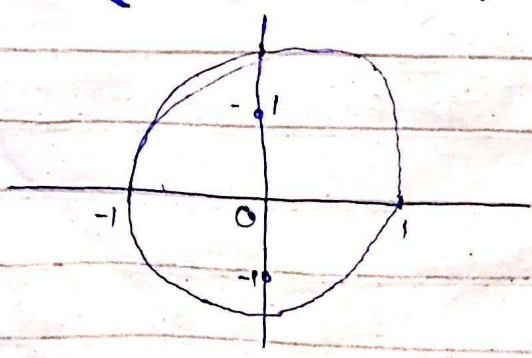
$$\frac{2x^2}{2}+\frac{y^2}{2}=1$$

$$x^2+\frac{y^2}{2}=1$$

$$c=\sqrt{a^2-b^2}=\sqrt{(1)^2-(\sqrt{2})^2}=\sqrt{1-2}=1$$

Foci $=(0,\pm c)=(0,\pm 1)$

Vertices $=(0\pm a)=(0\pm 1)$



(20) $2x^2+y^2=4$

$\frac{2x^2}{4}+\frac{y^2}{4}=0$

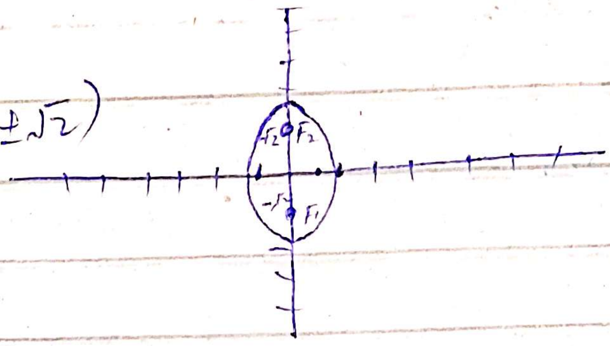
$\frac{x^2}{2}+\frac{y^2}{4}=0$

$c=\sqrt{a^2-b^2}=\sqrt{(2)^2-(\sqrt{2})^2}$

$=\sqrt{4-2}=\sqrt{2}$

$C=\sqrt{2}$

Foci $=(0,\pm\sqrt{2})$



(21) $3x^2+2y^2=6$

$\frac{3x^2}{6}+\frac{2y^2}{6}=1$

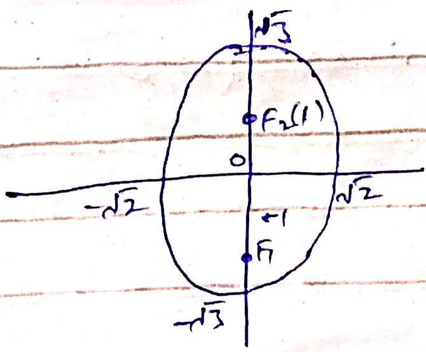
$\frac{x^2}{2}+\frac{y^2}{3}=1$

$c=\sqrt{a^2-b^2}=\sqrt{(\sqrt{3})^2-(\sqrt{2})^2}=\sqrt{3-2}=\sqrt{1}=1$

$C=1$

Foci $=(0,\pm 1)$

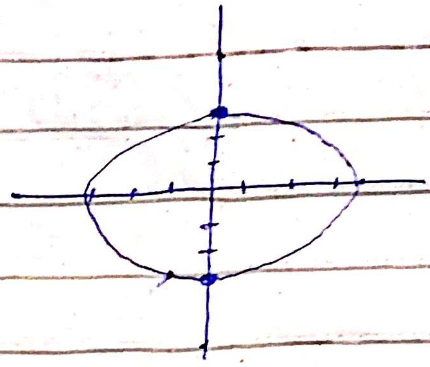
Vertices $=(0,\pm\sqrt{3})$



(22) $9x^2 + 10y^2 = 90$

$$\frac{9x^2}{90} + \frac{10y^2}{90} = 1$$

$$\frac{x^2}{10} + \frac{y^2}{9} = 1$$



$$c = \sqrt{a^2 - b^2} = \sqrt{(\sqrt{10})^2 - (3)^2} = \sqrt{10-9} = 1$$

Foci = $(\pm 1, 0)$

vertices = $(\pm a, 0) = (\pm 1, 0)$

origin, center $(0,0)$

(23) $6x^2 + 9y^2 = 54$

$$\frac{6x^2}{54_9} + \frac{9y^2}{54_6} = 1$$

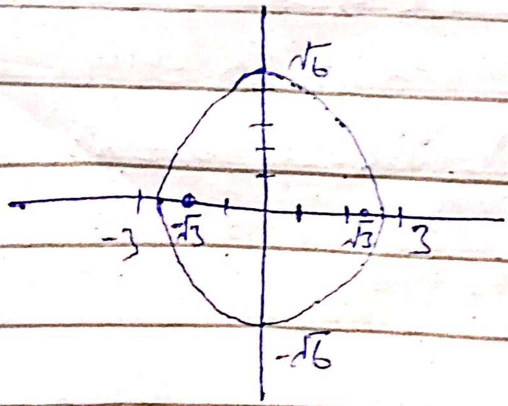
$$\frac{x^2}{9} + \frac{y^2}{6} = 1$$

$$c = \sqrt{a^2 - b^2} = \sqrt{(3)^2 - (\sqrt{6})^2} = \sqrt{9-6}$$

$c = \sqrt{3}$

Foci: $(\pm\sqrt{3}, 0)$

Vertices $(\pm 3, 0)$



(24) $169x^2 + 25y^2 = 4225$

$$\frac{169x^2}{4225} + \frac{25y^2}{4225} = 1$$

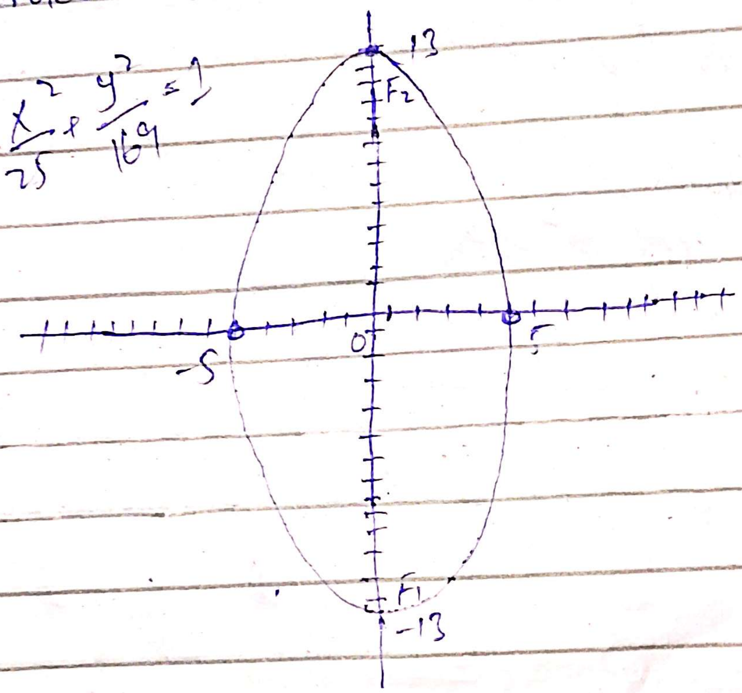
$$\frac{x^2}{25} + \frac{y}{169} = 1$$

$$c = \sqrt{a^2 - b^2} = \sqrt{(13)^2 - (5)^2} = \sqrt{169 - 25}$$

$C = \sqrt{144} = 12$

foci: $(0, \pm c) = (0, \pm 12)$

Vertices: $(0, \pm a) = (0, \pm 13)$



(25) Foci: $(\pm\sqrt{2}, 0)$ Vertices: $(\pm 2, 0)$

Find ellipse equation.

$a = 2$, $c = \sqrt{2}$, $b^2 = a^2 - c^2$

$b^2 = (2)^2 - (\sqrt{2})^2$

$b^2 = 4 - 2 = 2$

$b^2 = 2$

So the equation is

$$\frac{x^2}{4} + \frac{y^2}{2} = 1$$

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Foci: $(0, \pm 4)$

Vertices: $(0, \pm 5)$

$a = 5$, $c = 4$

$b^2 = (a)^2 - (c)^2 = (5)^2 - (4)^2 = 25 - 16$

$b^2 = 9$

Now

$$\boxed{\frac{x^2}{9} + \frac{y^2}{25} = 1}$$

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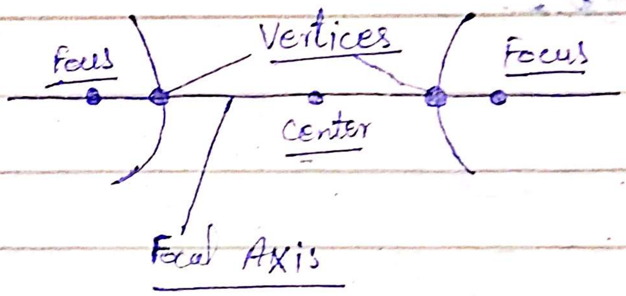
③ Hyperbolas:-

A hyperbola is the set of points in a plane whose distances from two fixed points in the plane have a constant difference. The two fixed points are the foci of the hyperbola.



Standard-Form equations for hyperbolas

⇒ Foci on the x-axis: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Center-to-focus distance

$$c = \sqrt{a^2+b^2}$$

Foci: $(\pm c, 0)$

Vertices: $(\pm a, 0)$

Asymptotes: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ or $y = \pm \frac{b}{a} x$

⇒ Foci on the y-axis:

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

Center-to-Focus distance: $c = \sqrt{a^2 + b^2}$

Foci: $(0, \pm c)$

Vertices: $(0, \pm a)$

Asymptotes: $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 0$ or $y = \pm \frac{a}{b} x$

Example:

$$\frac{x^2}{4} - \frac{y^2}{5} = 1$$

$a^2 = 4$, $b^2 = 5$

Center-to-Focus distance: $c = \sqrt{a^2 + b^2}$

$= \sqrt{(2)^2 + (\sqrt{5})^2}$

$= \sqrt{4+5} = \sqrt{9}$

$= 3$

Foci: $(\pm c, 0) = (\pm 3, 0)$

Vertices: $(\pm a, 0) = (\pm 2, 0)$

Center: $(0, 0)$

Asymptotes: $\frac{x^2}{4} - \frac{y^2}{5} = 0$ or $y = \pm \frac{\sqrt{5}}{2} x$

Example:- $x^2 - 4y^2 + 2x + 8y - 7 = 0$

$$x^2 + 2x - 4y^2 + 8y - 7 = 0$$

$$\left((x)^2 + 2(x)(1) + (1)^2 - (1)^2\right) - 4\left((y)^2 - 2(y)(1) + (1)^2 - (1)^2\right) = 7$$

$$(x+1)^2 - 4(y-1)^2 = 7 + 1 - 4$$

$$(x+1)^2 - 4(y-1)^2 = 4$$

$$\frac{(x+1)^2}{4} - \frac{4(y-1)^2}{4} = 1$$

$$\frac{(x+1)^2}{4} - (y-1)^2 = 1$$

$$\cancel{c^2 = \sqrt{a^2+b^2} = \sqrt{4-1}}$$

$$c^2 = a^2 + b^2 = 4 + 1 = 5$$

So the asymptotes are two lines

$$\frac{x+1}{2} - (y-1) = 0 \quad \text{and} \quad \frac{x+1}{2} + (y-1) = 0$$

or

$$y - 1 = \pm \frac{1}{2}(x+1)$$

The shifted foci have coordinates $(-1 \pm \sqrt{5}, 1)$.

(27) $2x^2 - 2y^2 = 4$

$$\frac{\cancel{2}x^2}{\cancel{4}_2} - \frac{\cancel{2}y^2}{\cancel{4}_2} = 1$$

$$\frac{x^2}{2} - \frac{y^2}{2} = 1$$

Center-to-Focus distance $= c = \sqrt{a^2 + b^2}$

$c = \sqrt{(\sqrt{2})^2 + (\sqrt{2})^2} = \sqrt{4} = 2$

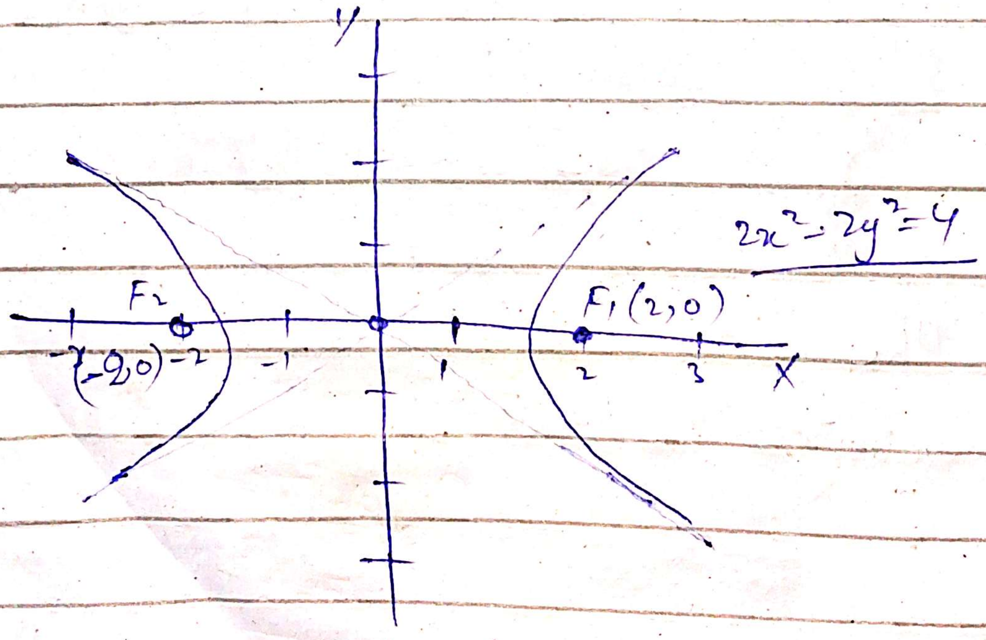
Foci $= (\pm c, 0) = (\pm 2, 0)$

Vertices $= (\pm a, 0) = (\pm 2, 0)$

Asymptotes $\frac{x^2}{2} - \frac{y^2}{2} = 0$ or $y = \pm \frac{b}{a} x$

$y = \pm \frac{2}{2} x$

$\boxed{y = \pm x}$



(28) $9x^2 - 16y^2 = 144$

$$\frac{\cancel{9}x^2}{\cancel{144}_{16}} - \frac{\cancel{16}y^2}{\cancel{144}_{9}} = 1$$

$$\frac{x^2}{16} - \frac{y^2}{9} = 1$$

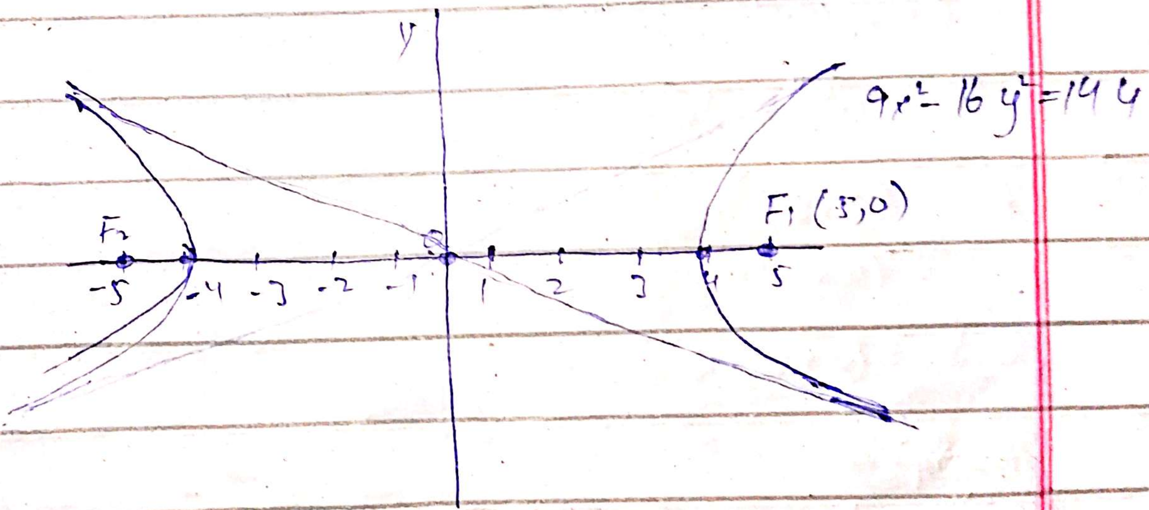
Center-to-focus distance : $c = \sqrt{a^2+b^2} = \sqrt{(4)^2+(3)^2}$

$c = \sqrt{16+9} = \sqrt{25} = 5$

Foci $= (\pm c, 0) = (\pm 5, 0)$

Vertices $= (\pm a, 0) = (\pm 4, 0)$

Asymptotes $\quad \frac{x^2}{4} - \frac{y^2}{3} = 0 \quad$ or $\quad y = \pm \frac{3}{4}x$



(29) $y^2 - x^2 = 10$

$$\frac{y^2}{10} - \frac{x^2}{10} = 1$$

Center - to - focus distance $= c = \sqrt{a^2+b^2}$

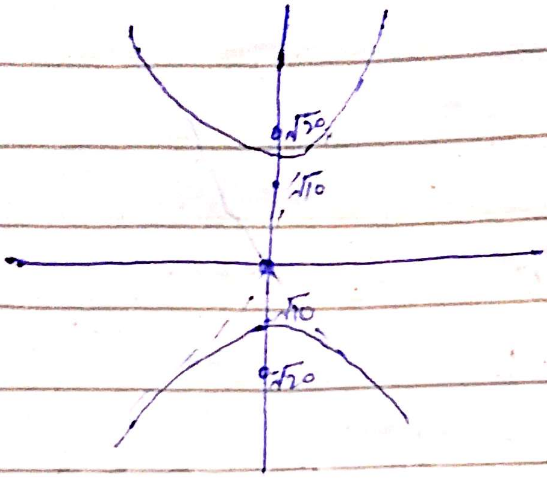
$c = \sqrt{(\sqrt{10}) + (\sqrt{10})} = \sqrt{20}$

Foci: $(\pm c, 0) = (\pm\sqrt{20}, 0)$

Vertices: $(\pm a, 0) = (\pm\sqrt{10}, 0)$

Asymptotes: $\frac{x^2}{\sqrt{10}} - \frac{y^2}{10} = 0$, or $y = \pm\frac{\sqrt{10}}{\sqrt{10}}x$

$$\boxed{y = \pm x}$$



(30) $y^2 - x^2 = 4$

$$\frac{y^2}{4} - \frac{x^2}{4} = 1$$

Center-to-focus distance

$$c = \sqrt{a^2+b^2} = \sqrt{4+4}$$

$$c = \sqrt{8} = 2\sqrt{2}$$

Foci: $(0, \pm c)$

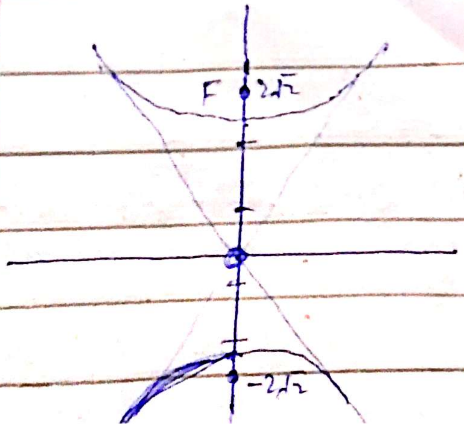
$= (0, \pm 2)$

Vertice $= (0, \pm a) = (0, \pm 2)$

Asymptotes

$\frac{y^2}{4} - \frac{x^2}{4} = 0$ or $y = \pm\frac{2}{2}x$

$y = \pm x$



(31)

$$6x^2 - 2y^2 = 12$$

$$\frac{6x^2}{12} - \frac{2y^2}{12_6} = 1$$

$$\frac{x^2}{2} - \frac{y^2}{6} = 1$$

$$c = \sqrt{a^2+b^2}$$

$$= \sqrt{(\sqrt{2})^2 + (\sqrt{6})^2}$$

$$= \sqrt{2+6} = \sqrt{8}$$

$$c = 2\sqrt{2}$$

Foci $= (\pm 2\sqrt{2}, 0)$

vertices $= (\pm\sqrt{2}, 0)$

Asymptotes

$$\frac{x^2}{2} - \frac{y^2}{6} = 0$$

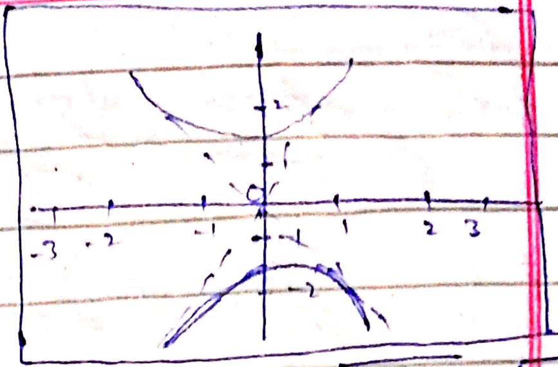
or $y = \pm\frac{\sqrt{6}}{\sqrt{2}}x$

$$\boxed{y = \pm\sqrt{3}x}$$

(37) $y^2 - 3x^2 = 3$

$$\frac{y^2}{3} - \frac{3x^2}{3} = 1$$

$$\frac{y^2}{3} - \frac{x^2}{1} = 1$$



Center-to-focus distance $= c = \sqrt{a^2+b^2} = \sqrt{(\sqrt{3})^2+(1)^2} = \sqrt{4}$

$$c = \sqrt{4} = 2$$

Foci : $(0, \pm 2)$

Asymptotes:-

$$\frac{y^2}{3} - x^2 = 0 \text{ or } y = \pm \frac{\sqrt{3}}{1} x$$

$$\boxed{y = \pm \sqrt{3}\, x}$$

(33) $6y^2 - 2x^2 = 12$

$2 | 24$
$2 | 12$
$2 | 6$
3
$2\sqrt{6}$

$$\frac{6y^2}{12} - \frac{2x^2}{12} = 1$$

$$y^2/2 - \frac{x^2}{6} = 1$$

Center-to-focus distance $= c = \sqrt{12+12} = \sqrt{24}$

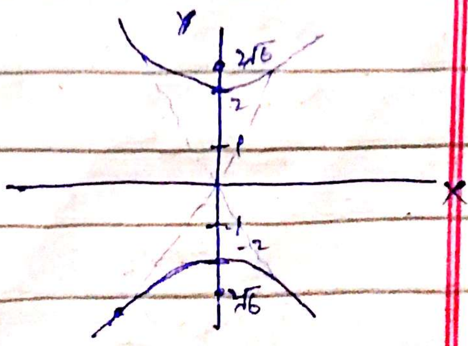
Foci $= (0, \pm\sqrt{24})$ $c = 2\sqrt{6}$

Vertices $= (0, \pm\sqrt{12}) = (0, \pm \cancel{\sqrt{12}}\ \sqrt{2})$

Asymptotes

$$\frac{y^2}{12} - \frac{x^2}{6} = 0, \text{ or } y = \pm \frac{\sqrt{2}}{\sqrt{6}} x$$

$$\boxed{y = \pm \frac{1}{\sqrt{3}} x}$$



(34) $64x^2 - 36y^2 = 2304$

$$\frac{64x^2}{2304} - \frac{36y^2}{2304} = 1$$

$$\frac{x^2}{36} - \frac{y^2}{64} = 1,$$

Center-to-focus distance $= c = \sqrt{a^2+b^2} = \sqrt{36+64}$

$c = \sqrt{100} = 10$

Foci: $(\pm c, 0) = (\pm 10, 0)$

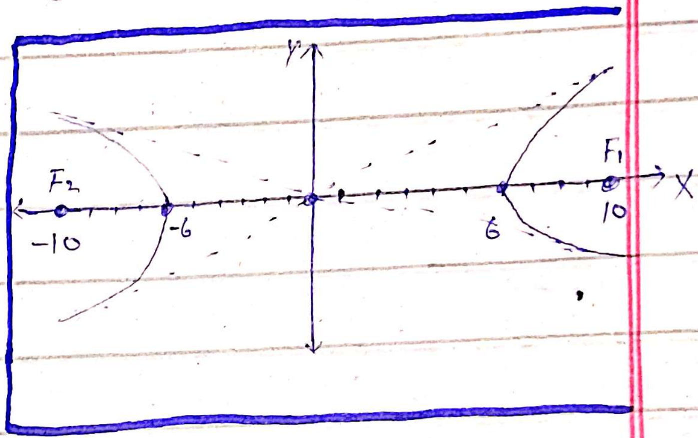
Vertices: $(\pm a, 0) = (\pm 6, 0)$

Asymptotes:-

$$\frac{x^2}{6} - \frac{y^2}{8} = 0$$

or $y = \pm \frac{8}{6}x$

$$\boxed{y = \pm \frac{4}{3}x}$$



(35) Foci: $(0, \pm\sqrt{2})$, Asymptotes $y = \pm x$

Find hyperbola's standard-form equation

$c = \sqrt{2}$ and $\frac{a}{b} = 1$

$a = b$

$c^2 = a^2 + b^2$

$c^2 = 2a^2$

$(\sqrt{2})^2 = 2a^2$

$\cancel{2} = \cancel{2}a^2$

$a^2 = 1$

$\boxed{a = 1}$

$\boxed{b = 1}$

Then eq.

$$\boxed{y^2 - x^2 = 1}$$

(36) Foci: $(\pm 2, 0)$, Asymptotes: $y = \pm \frac{1}{\sqrt{3}} x$

* Find eq. of hyperbola

$\Rightarrow c = 2.$, $\Rightarrow \frac{b}{a} = \frac{1}{\sqrt{3}} \Rightarrow b = \frac{a}{\sqrt{3}}$

$= c^2 = a^2 + b^2$

$c^2 = a^2 + \left(\frac{a}{\sqrt{3}}\right)^2 = a^2 + \frac{a^2}{3} = \frac{4a^2}{3}$

$(2)^2 = \frac{4a^2}{3}$

$(\cancel{4})(3) = \cancel{4}a^2$

$\boxed{a = \sqrt{3}}$

$b = \frac{a}{\sqrt{3}} = \frac{\sqrt{3}}{\sqrt{3}} = \boxed{1}$

$\boxed{\frac{x^2}{\sqrt{3}} - y^2 = 1}$

(37) Vertices: $(\pm 3, 0)$

Asymptotes: $y = \pm \frac{4}{3} x$, Find eq.

$\boxed{a = 3}$ ✓

$\frac{b}{a} = \frac{4}{3}$

$b = \left(\frac{4}{3}\right) a = \left(\frac{4}{\cancel{3}}\right)(\cancel{3}) = \boxed{4}$ ✓

$\boxed{\frac{x^2}{9} - \frac{y^2}{16} = 1}$

(38) Vertices : $(0, \pm 2)$
Asymptotes : $y = \pm \frac{1}{2}x$. Find eq.

$\boxed{a = 2}$

$\frac{a}{b} = \frac{1}{2}$

$b = 2a = 2(2) = \boxed{4}$

$\frac{y^2}{(a)^2} - \frac{x^2}{(b)^2} = 1$

$$\boxed{\frac{y^2}{4} - \frac{x^2}{16} = 1}$$

* Conics in Polar coordinates

Formula

$$r = \frac{ep}{1 \pm e\cos\theta} \quad \text{or} \quad r = \frac{ep}{1 \pm e\sin\theta}$$

where

- r is the distance from the origin (focus)
- e is the ecentricity of the conic
- p is the distance from the focus to the directrix.

$\Rightarrow$ Parabola: $e = 1 = \frac{c}{a} = \cancel{}$ 1

$\Rightarrow$ Ellipse: $e < 1 = \frac{c}{a} = \frac{\sqrt{a^2-b^2}}{a}$

$\Rightarrow$ Hyperbola: $e > 1e > 1 = \frac{c}{a} = \frac{\sqrt{a^2+b^2}}{a}$

$\boxed{x = 0 \pm \frac{a}{e}}$

$\Rightarrow$ Equations of Parabola

- $r = \frac{p}{1+\cos\theta}$ or $r = \frac{p}{1-\cos\theta}$
- $r = \frac{p}{1+\sin\theta}$ or $r = \frac{p}{1-\sin\theta}$

$\Rightarrow$ Equations of Ellipse

- $r = \frac{a(1-e^2)}{1+e\cos\theta}$ or $\frac{a(1-e^2)}{1-e\cos\theta}$
- $r = \frac{a(1-e^2)}{1+e\sin\theta}$ or $r = \frac{a(1-e^2)}{1-e\sin\theta}$

⇒ Equations of Hyperbola

- $r = \dfrac{a(e^2-1)}{1+e\cos\theta}$ or $r = \dfrac{a(e^2-1)}{1-e\cos\theta}$

- $r = \dfrac{a(e^2-1)}{1+e\sin\theta}$ or $r = \dfrac{a(e^2-1)}{1-e\sin\theta}$

~~Equ~~

Examples:-

Eccentricity of the ellipse,

$16x^2+25y^2=400$

$\dfrac{16x^2}{400_{25}} + \dfrac{25y^2}{400_{16}} = 1$

$\dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$

$a^2 = 25$ | $b^2 = 16$

$a = 5$ | $b = 4$

$e = \dfrac{c}{a} = \dfrac{\sqrt{a^2-b^2}}{a}$

$e = \dfrac{\sqrt{25-16}}{5}$

$\boxed{e = \dfrac{\sqrt{9}}{5}}$ $\boxed{= \frac{3}{5}}$

$c = \sqrt{a^2-b^2} = \sqrt{25-16} = \sqrt{9} = 3$

Foci: $\cancel{(\pm\sqrt{41}, 0)}$ $(\pm 3, 0)$

directrices are $x = 0 \pm \dfrac{a}{e}$

$= 0 \pm \dfrac{5}{3/5} = \pm \dfrac{25}{3}$

(3) $2x^2+y^2=2$

$$\frac{\cancel{2}x^2}{\cancel{2}}+\frac{y^2}{2}=1$$

$$x^2+\frac{y^2}{2}=1$$

$$e=\sqrt{a^2-b^2}=\sqrt{(2)^2-(1)}=\sqrt{1}=1$$

$$e=\frac{c}{a}=\frac{1}{\sqrt{2}}$$

$F(0, \pm 1)$

directrices are $x = 0 \pm \frac{a}{e}$

$$= 0 \pm \frac{\sqrt{2}}{1/\sqrt{2}} = \boxed{\pm 2}$$

(3) $169x^2+25y^2=4225$

$$\frac{169}{\underset{25}{4225}}x^2+\frac{25}{\underset{169}{4225}}y^2=1$$

$$\frac{x^2}{25}+\frac{y^2}{169}=1$$

$$c=\sqrt{a^2-b^2}=\sqrt{169-25}=\sqrt{144}=12$$

$$e=\frac{c}{a}=\frac{12}{13}$$

$F(0, \pm 13)$

directrices are $x = 0 \pm \frac{a}{e} = 0 \pm \frac{13}{12/13}$

$$= \boxed{\pm \frac{169}{12}}$$

(9) Foci: $(0, \pm 3)$, $e = 0.5$ find ellipse eq.

$c = 3$

$$a = \frac{c}{e} = \frac{3}{0.5} = 6$$

$$b^2 = a^2 - c^2$$

$$b^2 = 36 - 9 = 27$$

$$\frac{x^2}{27} + \frac{y^2}{36} = 1$$

(11) Vertices: $(0, \pm 70)$, $e = 0.1$ Find ellipse eq.

$$a = 70$$

$$a = \frac{c}{e} = \frac{c}{0.1} =$$

$$70 = \frac{c}{0.1}$$

$$c = 70(0.1) = 7$$

$$b^2 = a^2 - c^2$$

$$b^2 = 4900 - 49 = 4851$$

Then

$$\frac{x^2}{4851} + \frac{y^2}{4900} = 1$$

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(13) Focus: $(\sqrt{5}, 0)$; directrix: $x = \frac{9}{\sqrt{5}}$

Find ellipse eq.

$c = \sqrt{5}$

$$\frac{a}{e} = \frac{9}{\sqrt{5}} = \frac{ae}{e^2} = \frac{9}{\sqrt{5}}$$

$$= \frac{\sqrt{5}}{e^2} = \frac{9}{\sqrt{5}}$$

$$e^2 = \frac{5}{9}$$

$$e = \frac{\sqrt{5}}{3}$$

Then $PF = \frac{\sqrt{5}}{3} PD = \sqrt{(x-\sqrt{5})^2 + (y-0)^2}$

$$= \frac{\sqrt{5}}{3}\left| x - \frac{9}{\sqrt{5}} \right|$$

$$= (x-\sqrt{5})^2 + y^2 = \frac{5}{9}\left(x - \frac{9}{\sqrt{5}}\right)^2$$

$$x^2 - 2\sqrt{5}x + 5 + y^2 = \frac{5}{9}\left(x^2 - \frac{18}{\sqrt{5}}x + \frac{81}{5}\right)$$

$$= \frac{4}{9}x^2 + y^2 = 4$$

$$\boxed{\frac{x^2}{9} + \frac{y^2}{4} = 1}$$

Hyperbolas and Eccentricity

17 $x^2 - y^2 = 1$

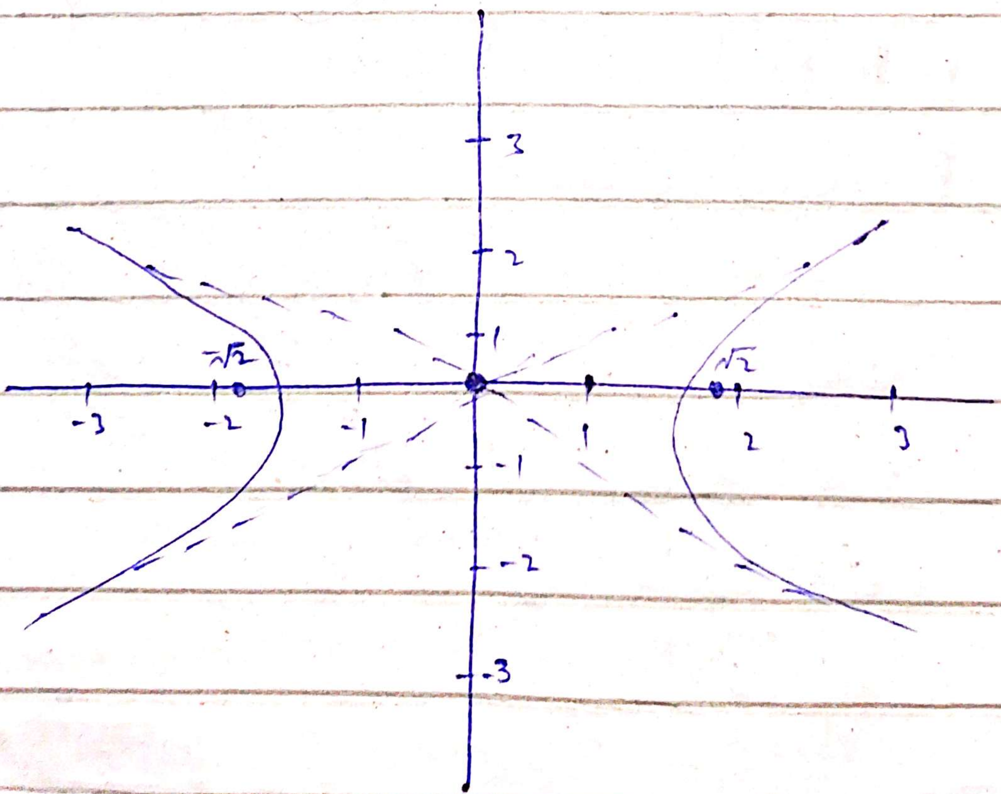
$c = \sqrt{a^2+b^2} = \sqrt{1^2+1^2} = \sqrt{2}$

$e = \frac{c}{a} = \boxed{\sqrt{2}}$

Asymptotes $\boxed{y = \pm x}$

Foci : $(\pm\sqrt{2}, 0)$

directrice are $x = 0 \pm \frac{a}{e} = \pm \frac{1}{\sqrt{2}}$



(24) $64x^2 - 36y^2 = 2304$

$$\frac{\cancel{64}x^2}{\cancel{2304}_{36}} - \frac{\cancel{36}y^2}{\cancel{2304}_{64}} = 1$$

$$\frac{x^2}{36} - \frac{y^2}{64} = 1$$

$$c = \sqrt{a^2+b^2} = \sqrt{\cancel{3}64+36} = \sqrt{100} = 10$$

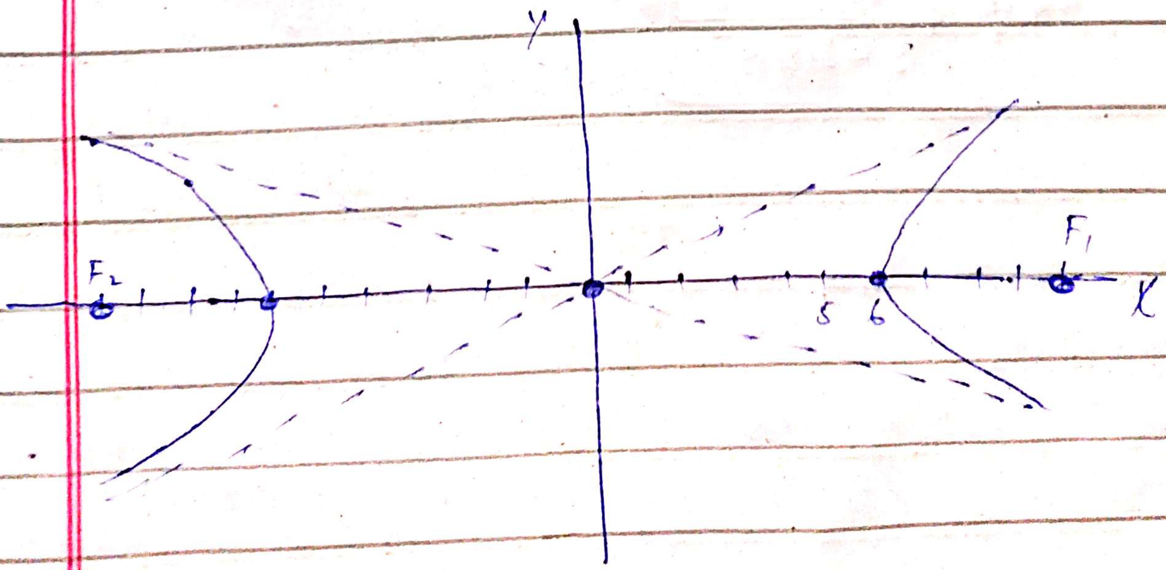
$$e = \frac{c}{a} = \frac{\cancel{10}}{\cancel{6}} = \boxed{\frac{5}{3}}$$

Foci: $\boxed{(10, 0)}$

directrices $x = 0 \pm \frac{a}{e} = \pm \frac{6}{5/3} = \pm \frac{18}{5}$

Asymptotes

$$\boxed{y = \pm \frac{4}{3}x}$$



(29) $e = 1$, $x = 2$

Find a polar equation of each conic section

$e = 1$, $x = P = 2$

The formula is

$$r = \frac{eP}{1+e\cos\theta} = \frac{(1)(2)}{1+(1)\cos\theta}$$

$$\boxed{r = \frac{2}{1+\cos\theta}}$$

(36) $e = 1/3$, $y = 6$. Find Polar equation

$$r = \frac{eP}{1+e\sin\theta} = \frac{(1/3)(6)}{1+(1/3)\sin\theta}$$

$$\boxed{r = \frac{6}{3+\sin\theta}}$$

(45) $r\cos\left(\theta - \frac{\pi}{4}\right) = \sqrt{2}$ Find cartesian equation.

$$r\cos\left(\theta - \frac{\pi}{4}\right) = \sqrt{2}$$

$$r\left(\cos\theta \cos\frac{\pi}{4} + \sin\theta \sin\frac{\pi}{4}\right) = \sqrt{2}$$

$$r\left(\frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta\right) = \sqrt{2}$$

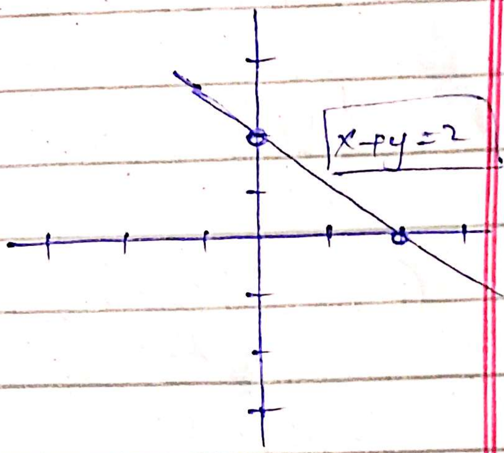
$$\frac{1}{\sqrt{2}} r\cos\theta + \frac{1}{\sqrt{2}} r\sin\theta = \sqrt{2}$$

$$\frac{1}{\sqrt{2}} x + \frac{1}{\sqrt{2}} y = \sqrt{2}$$

$$\frac{x+y}{\sqrt{2}} = \sqrt{2}$$

$$x+y = 2$$

$$\boxed{y = 2 - x}$$



(48) $r\cos\left(\theta + \frac{\pi}{3}\right) = 2$ Find cartesian equation.

$$r\cos\left(\theta + \frac{\pi}{3}\right) = 2$$

$$r\left(\cos\theta \cos\frac{\pi}{3} - \sin\theta \sin\frac{\pi}{3}\right) = 2$$

$$r\left(\cos\theta\left(\frac{1}{2}\right) - \sin\left(\frac{\sqrt{3}}{2}\right)\right) = 2$$

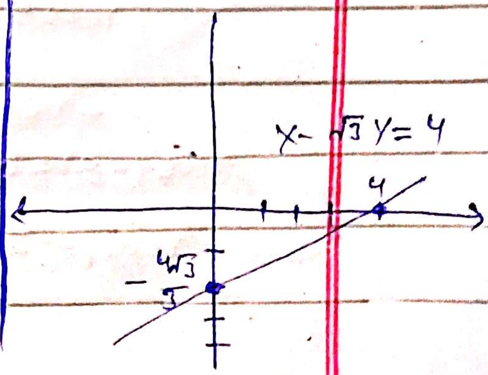
$$\frac{1}{2} r\cos\theta - \frac{\sqrt{3}}{2} r\sin\theta = 2$$

$$\frac{1}{2}x - \frac{\sqrt{3}}{2}y = 2$$

$$\frac{x - \sqrt{3}y}{2} = 2$$

$$x - \sqrt{3}y = 4$$

$$y = \frac{\sqrt{3}}{3}x - \frac{4\sqrt{3}}{3}$$



(49) $\sqrt{2}x + \sqrt{2}y = 6$ Find polar equation.

$\sqrt{2}\ r\cos\theta + \sqrt{2}\ r\sin\theta = 6$

$r(\sqrt{2}\cos\theta + \sqrt{2}\sin\theta) = 6$

$r\left(\frac{\sqrt{2}}{2}\cos\theta + \frac{\sqrt{2}}{2}\sin\theta\right) = 3$

$r\left(\frac{1}{\sqrt{2}}\cos\theta + \frac{1}{\sqrt{2}}\sin\theta\right) = 3$

$r\left(\cos\frac{\pi}{4}\cos\theta + \sin\frac{\pi}{4}\sin\theta\right) = 3$

$\boxed{r\cos\left(\theta - \frac{\pi}{4}\right) = 3}$

(50) $y = -3$ Find polar equation.

$r\sin\theta = -3$

$-r\sin\theta = 3$

$r\sin(-\theta) = 3$

$\cancel{r\sin\left(\frac{\pi}{2}\right.}$

$r\cos\left(\frac{\pi}{2} - (-\theta)\right) = 3$

$\boxed{r\cos\left(\frac{\pi}{2} + \theta\right) = 3}$

(52) $x = -6$

$r\cos\theta = -6$

$-r\cos\theta = 6$

$r\cos(-\theta) = 6$

$r\cos(\theta - \pi) = 4$

Angle between two lines

The angle between two lines can be calculated by formula

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

where
- θ is the angle betwee two lines
- m_1 and m_2 are slopes of lines

* For Parallel lines $m_1 = m_2$

* For perpendicular lines $m_1 m_2 = -1$

Questions:-

$y = 2x + 3$, $y = x - 2$

Find angle between these lines

Solution:-

The slope m_1 is 2 and $m_2 = 1$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{2 - 1}{1 + (2)(1)}\right| = \left|\frac{1}{1+2}\right|$$

$$\tan\theta = \left|\frac{1}{3}\right| = \frac{1}{3}$$

$$\theta = \tan^{-1}\left(\frac{1}{3}\right) \approx 18.43^\circ$$

$$\boxed{\theta \approx 18.43^\circ}$$

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Q. 2 Find angle between the lines $y = 3x - 2$ and $y = 2x + 1$.

Solve:-

$m_1 = 3$ and $m_2 = 2$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{3-2}{1+(3)(2)}\right|$$

$$\tan\theta = \left|\frac{1}{1+6}\right| = \left|\frac{1}{7}\right| = \boxed{\frac{1}{7}}$$

$$\theta = \tan^{-1}\left(\frac{1}{7}\right) \approx \boxed{8.13^\circ}$$

Q3 $y = -2x + 4$, $y = x + 3$

Solve:-

$m_1 = -2$, $m_2 = 1$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{-2-1}{1+(-2)(1)}\right| = \left|\frac{-3}{1-2}\right|$$

$$\tan\theta = \left|\frac{-3}{-1}\right| = 3$$

$$\theta = \tan^{-1}(3) \approx \boxed{71.57}$$

Equation of Locus

The eq. Locus is the eq. that represents the set of all points that satisfy certain given conditions.

Questions.

(Q1) Find equation of locus of points that are equidistant from the points (2,3) and (4,5)

Solution:-

Let (x,y) be a point on the locus. Then, the distance from (x,y) to $(2,3)$ is equal to the distance from (x,y) to $(4,5)$.

By using distance formula.

$$\sqrt{(x-2)^2+(y-3)^2} = \sqrt{(x-4)^2+(y-5)^2}$$

Squaring both sides

$$(x-2)^2+(y-3)^2 = (x-4)^2+(y-5)^2$$

$$x^2-4x+4+y^2-6y+9 = x^2-8x+16+y^2-10y+25$$

$$\cancel{x^2}-\cancel{x^2}-4x+8x+\cancel{y^2}-\cancel{y^2}-6y+10y+4+9-16-25=0$$

$$4x+4y-28=0$$

$$4(x+y-7)=0$$

$$\boxed{x+y-7=0}$$

Q3 Find the equation of locus of points that are 3 units from the point (1,2)

Let (x,y) be a point on locus:

Using distance formula

$$\sqrt{(x-1)^2+(y-2)^2} = 3$$

Squaring both sides

$$(x-1)^2+(y-2)^2 = 9$$

$$x^2 - 2x + 1 + y^2 - 4y + 4 = 9$$

$$\boxed{x^2 - 2x + y^2 - 4y - 4 = 0}$$

Shortest distance between two lines

Formula

$$d = \frac{|\vec{b_1} \times \vec{b_2} \cdot (\vec{a_2} - \vec{a_1})|}{|\vec{b_1} \times \vec{b_2}|}$$

$\boxed{Q.1}$ Line 1: $x = 2t+1,\ y = 3t-2,\ z = t+4$

Line 2: $x = s+2,\ y = 2s-1,\ z = 3s+1$

Solution:- The direction vectors of lines are $\vec{b_1} = (2, 3, 1)$ and $\vec{b_2} = (1, 2, 3)$.
A point on each line is $(1, -2, 4)$ and $(2, -1, 1)$ respectively.

$$\vec{a_2} - \vec{a_1} = (1, 1, -3)$$

$$\vec{b_1} \times \vec{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{vmatrix} = \hat{i}\begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} - \hat{j}\begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} + \hat{k}\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix}$$

$$= \hat{i}(9-2) - \hat{j}(6-1) + \hat{k}(4-3)$$

$$= 7\hat{i} - 5\hat{j} + \hat{k}$$

$$\vec{b_1} \times \vec{b_2} = (7, -5, 1)$$

The shortest distance is

$$d = \frac{|(7, -5, 1) \cdot (1, 1, -3)|}{\sqrt{7^2 + (-5)^2 + 1^2}} = \frac{|-15|}{\sqrt{75}} = \sqrt{3}$$

(2) Line 1: $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z+2}{1}$

Line 2: $\frac{x+2}{1} = \frac{y-2}{2} = \frac{z+1}{3}$

Sol:-

$\vec{b_1} = (2, 3, 1)$ and $\vec{b_2} = (1, 2, 3)$

$\vec{a_1} = (1, -1, -2)$. $\vec{a_2} = (-2, 2, -1)$

$\vec{a_2} - \vec{a_1} = (3, -3, 3)$

$$\vec{b_1} \times \vec{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{vmatrix}$$

$$= \hat{i}\begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} - \hat{j}\begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} + \hat{k}\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix}$$

$$= \hat{i}(9-2) - \hat{j}(6-1) + \hat{k}(4-3)$$

$\vec{b_1} \times \vec{b_2} = 7\hat{i} - 5\hat{j} + \hat{k}$

$\vec{b_1} \times \vec{b_2} = (7, -5, 1)$

Now formula

$$d = \frac{|(7, -5, 1) \cdot (3, -3, 3)|}{\sqrt{7^2 + (-5)^2 + (1)^2}} = \frac{|-39|}{\sqrt{75}}$$

$$= \frac{13\sqrt{3}}{5}$$

Area of Surface of revolution

The area of a surface area of a solid formed by rotating a curve about an axis.

Formula:-

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$$

or

$$S = 2\pi \int_a^b x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$$

Questions

Q.1 Find the area of the surface of revolution formed by rotating the curve $y = x^2$ about the X-axis from $x = 0$ to $x = 1$.

Solution:-

Using formula

$$S = 2\pi \int_0^1 x^2 \sqrt{(2x)^2}\, dx$$

Evaluating the integral, we get

$$\boxed{S = \frac{\pi}{6}\left(5\sqrt{5} - 1\right)}$$

Q.2 Find the area of the surfaces of revolution formed by rotating the curve $y = \sqrt{x}$ about the x-axis from $x=0$ to $x=4$.

Solution:-

$$S = 2\pi \int_0^4 \sqrt{x} \sqrt{1 + \left(\frac{1}{2\sqrt{x}}\right)^2}\, dx$$

$$S = 2\pi \int_0^4 \sqrt{x} \sqrt{1 + \frac{1}{4x}}\, dx$$

by solving integral we have

$$S = \frac{\pi}{6}(17\sqrt{17} - 1)$$